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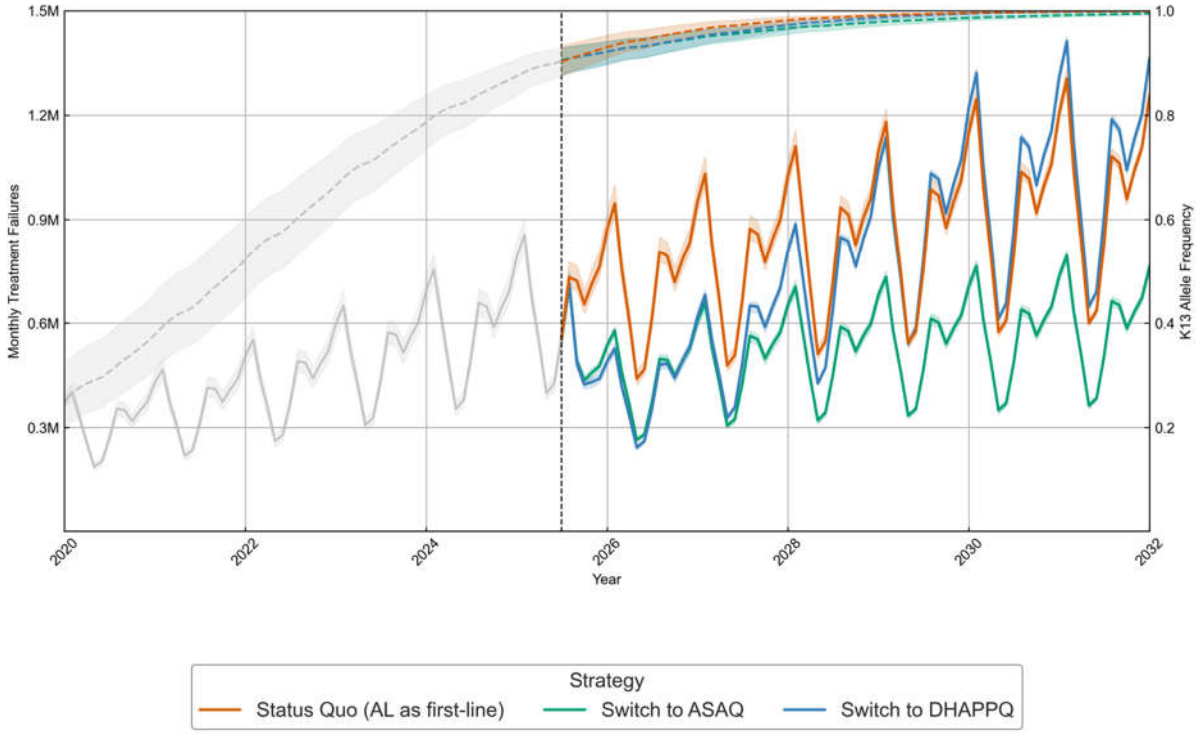

Figure 1. Trends in malaria treatment failures and *pfkelch13* allele frequency (2020–2032) under the status quo strategy in Uganda of continued artemether lumefantrine use (burnt orange), a switch to artesunate-amodiaquine (green), and a switch to dihydroartemisinin-piperaquine (blue). Gray lines show the model burn-in and calibration prior to July 2025. Solid lines show monthly treatment failure counts (left axis) while dashed lines show the combined *pfkelch13* allele frequency (sum of frequencies for the 469Y and 675V alleles) which is estimated to be near 0.90 in July 2025. The monthly treatment failure count includes all projected treatment failures for all antimalarials used in the general population. Lines show medians while shaded regions denote the 90% ranges across simulation outcomes (N = 32).

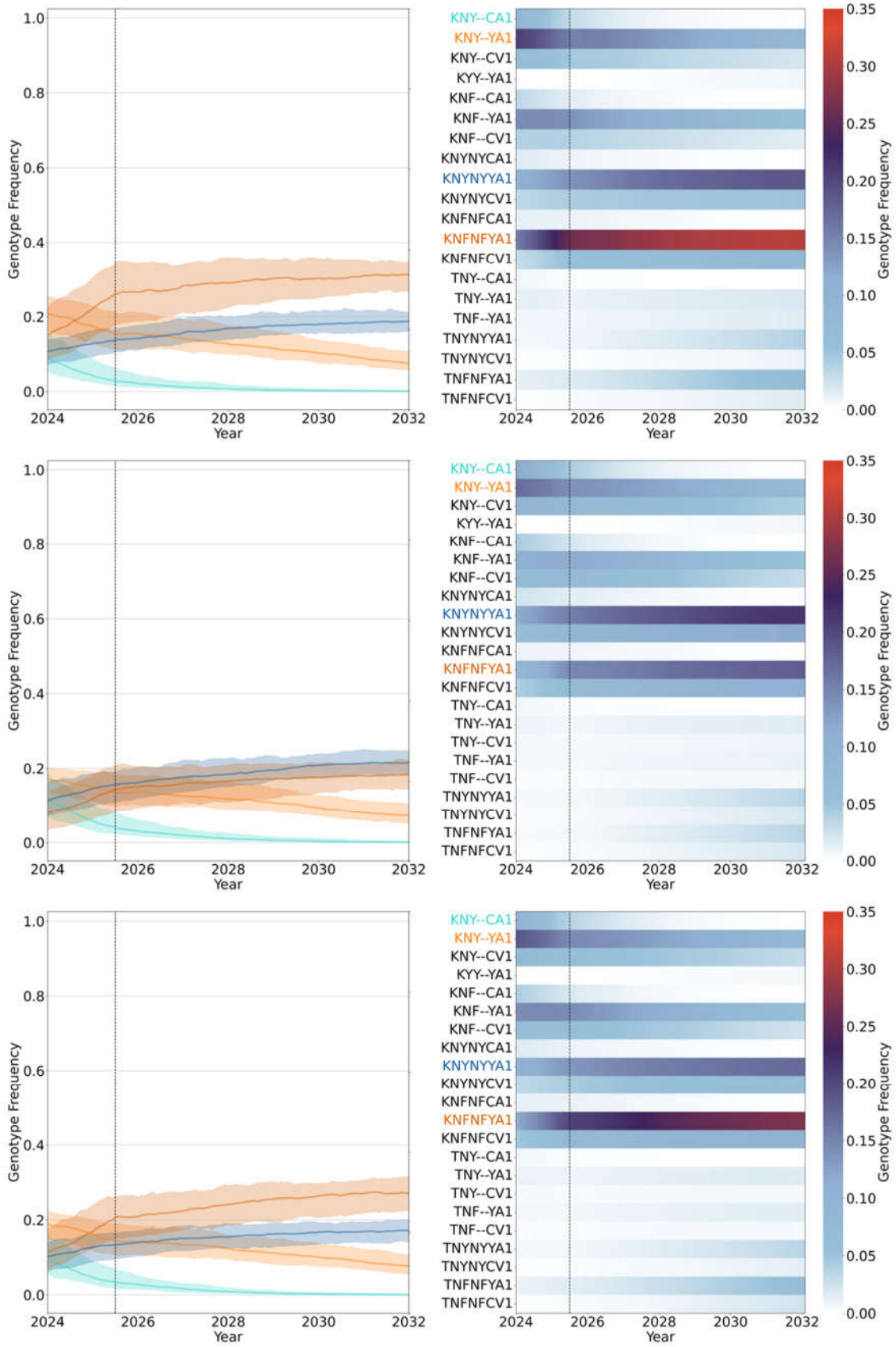
Genotype Frequency
Year
KNY--CA1
KNY--YA1
KNY--CV1
KYY--YA1
KNF--CA1
KNF--YA1
KNF--CV1
KNYNYCA1
KNYNYYA1
KNYNYCV1
KNFNFCA1
KNFNFYA1
KNFNFCV1
TNY--CA1
TNY--YA1
TNY--CV1
TNF--YA1
TNF--CV1
TNYNYYA1
TNYNYCV1
TNFNFYA1
TNFNFCV1

Figure 2. Three separate stochastic simulations of genotype evolution under deployment of artesunate-amodiaquine from 2025 to 2032. Left panels show genotype frequencies of three common genotypes (blue, orange, burnt orange) with artemisinin/lumefantrine double-resistance and one common genotype (light green) carrying mutations associated with lumefantrine resistance. The burnt orange and blue genotypes result in higher treatment failure (with AL use) due to an additional copy of *pfmdr1* in these parasites, and the higher treatment failure rate allows these two genotypes to outcompete the orange genotype (KNY--YA1). Note that ASAQ is deployed in the public sector only and 41.2% of all treatments are private-sector AL use; this results in significant selection pressure from AL during the entire simulation period. Lines show medians while shaded area shows 90% range of genotype frequency across all districts of Uganda. Genotype evolution heatmaps for each of the three stochastic simulations are shown in the right-hand column, with each row showing the rise and fall of a particular genotype from 2024 to 2032. Genotype codes (from left to right) are based on the following loci: *pfcrt* K76T, *pfmdr1* N86Y, *pfmdr1* Y184F, *pfmdr1* presence/absence of second copy, *pfkelch13* C469Y, *pfkelch13* A675V, copy number of plasmepsin genes. Genotypes with early resistance to amodiaquine would show a T in the first position of the genotype code (bottom 7 or 9 rows in each heatmap) but none of these rises to a genotype frequency above 0.12. In 2032, *pfcrt* 76T allele frequencies are 0.15, 0.18, and 0.18 (top to bottom) in the three simulations. All simulated genotypes rising above 0.005 frequency are included in the heatmaps on the right-hand side.

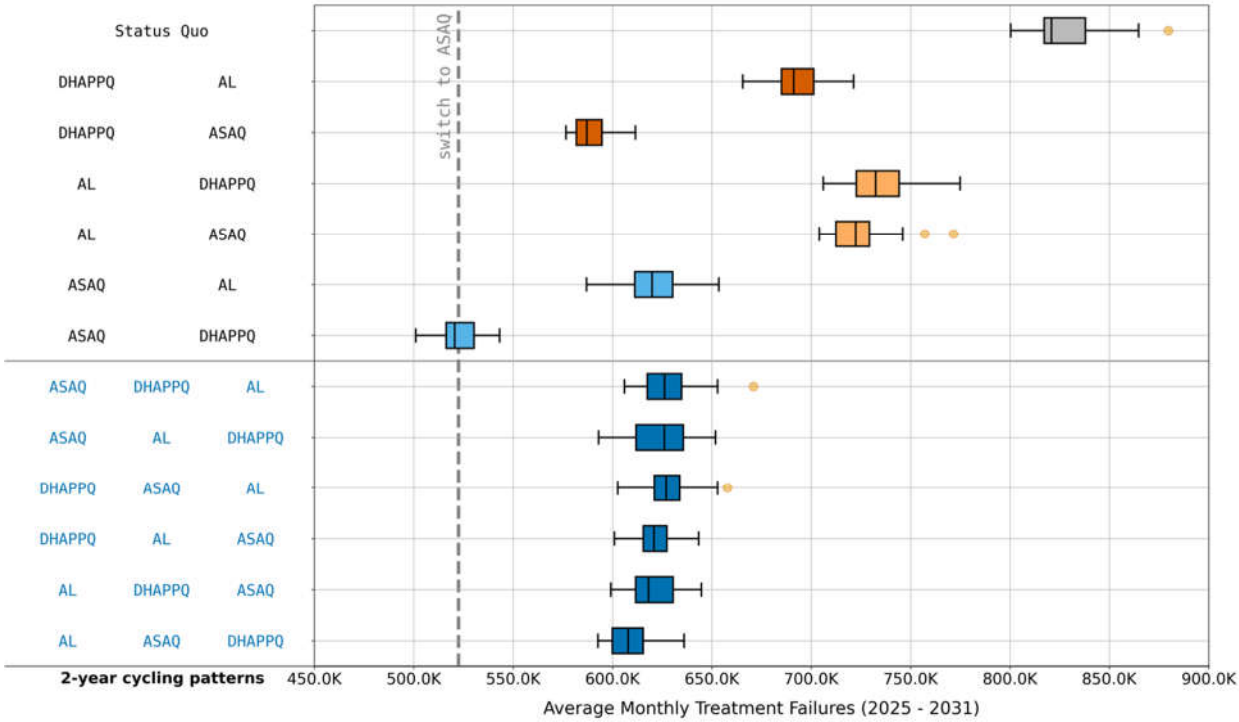

Figure 3. Average monthly treatment failures (2025 - 2031) under different 2-year cycling strategies. These boxplots, summarizing N = 32 simulations, illustrate the projected average monthly treatment failures for various 2-year cycling regimens using artemether-lumefantrine (AL), artesunate-amodiaquine (ASAQ), and dihydroartemisinin-piperaquine (DHA-PPQ). The vertical dashed line represents the median monthly treatment failures for a single-drug switch to ASAQ after July 2025. Sky blue boxplots correspond to two-drug cycles starting with ASAQ, red boxplots represent two-drug cycles initiated with DHA-PPQ, orange boxplots indicate two-drug cycles starting with AL, and dark blue boxplots represent three-drug cycling strategies. While three-therapy cycling strategies achieve low treatment failure counts, none of these outperforms a two-therapy cycle of ASAQ followed by DHA-PPQ. Boxplot centers are medians, box bounds show interquartile range, and whiskers show 1.5 IQR.

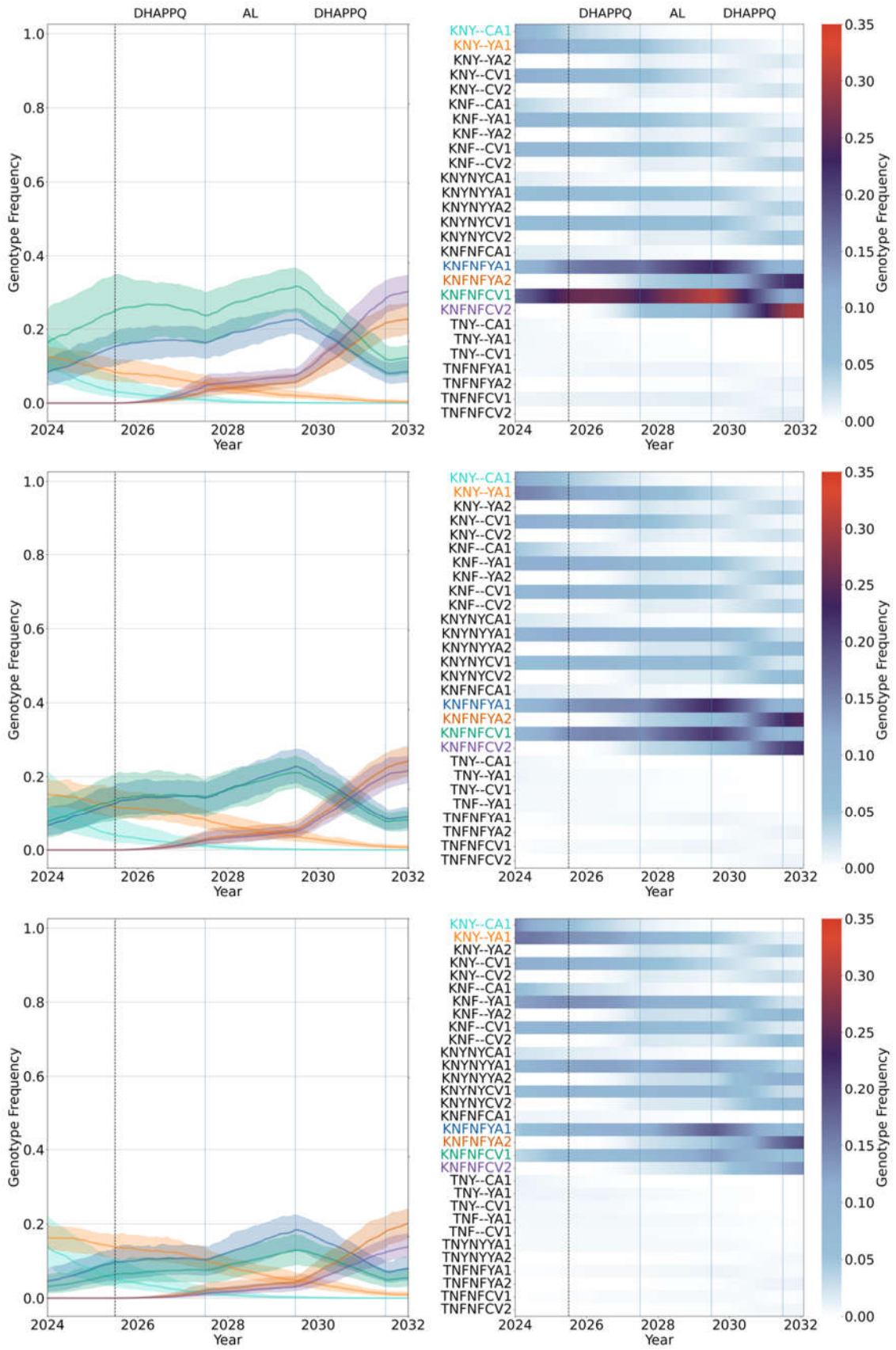
DHAPPQ
AL
DHAPPQ
Genotype Frequency
Year
KNY--CA1
KNY--YA1
KNY--YA2
KNY--CV1
KNY--CV2
KNF--CA1
KNF--YA1
KNF--YA2
KNF--CV1
KNF--CV2
KNYNYCA1
KNYNYYA1
KNYNYYA2
KNYNYCV1
KNYNYCV2
KNFNFCA1
KNFNFYA1
KNFNFYA2
KNFNFCV1
KNFNFCV2
TNY--CA1
TNY--YA1
TNY--CV1
TNF--YA1
TNF--CV1
TNYNYYA1
TNYNYYA2
TNFNFYA1
TNFNFYA2
TNFNFCV1
TNFNFCV2

Figure 4. Three separate stochastic simulations of genotype evolution under a 2-year cycling strategy with DHA-PPQ used first, followed by 2 years of AL (in both the public and private sector), and then followed by 2 years of DHA-PPQ use (public sector only). AL is used in the private sector throughout. Left panels show genotype frequencies of six common genotypes: five of these carry a *pfkelch13* mutation, and two of these genotypes (burnt orange, purple) carry triple resistance to lumefantrine, piperaquine, and artemisinin. In these simulations, we see that double-resistance to AL (blue, green) continues to be selected for, from 2026 to 2029, whether DHA-PPQ is used in the public sector or not. From mid-2027 to mid-2029, when AL is used in both public and private sectors, double-resistant genotypes to AL are selected for, and the KNFNFYA1 (blue) and KNFNFCV1 (green) genotypes remain at high levels. When DHA-PPQ is deployed for the second time in July 2029, the two triple-resistant genotypes (KNFNFYA2 and KNFNFCV2) are selected for quickly because PPQ resistance has a large fitness advantage (high treatment failure rate) over PPQ sensitivity. Lines show medians while shaded areas in left panels show 90% range of genotype frequency across all districts of Uganda. Genotype evolution heatmaps for each of the three stochastic simulations are shown in the right-hand column, with each row showing the rise and fall of a particular genotype from 2024 to 2032. Genotype codes (from left to right) are based on the following loci: *pfcrt* K76T, *pfmdr1* N86Y, *pfmdr1* Y184F, *pfmdr1* presence/absence of second copy, *pfkelch13* C469Y, *pfkelch13* A675V, copy number of plasmepsin genes. All simulated genotypes rising above 0.005 frequency are included in the heatmaps on the right-hand side.

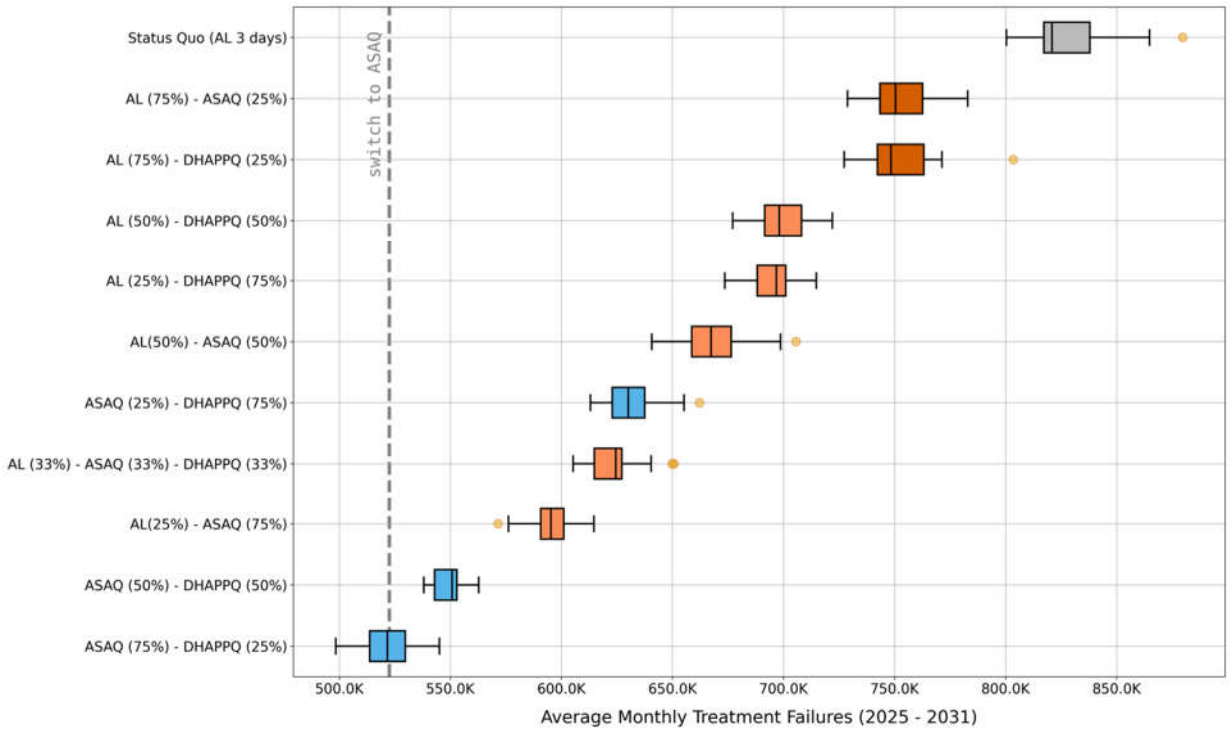

Figure 5. Average monthly treatment failures (2025-2031) across different MFT strategies. Each boxplot summarizes N = 32 simulation outcomes and shows the average monthly treatment failure count, from 2025 to 2031, under various multiple first-line therapies (MFT) strategies. Strategies differ according to the different proportions of artemether-lumefantrine (AL), artesunate-amodiaquine (ASAQ), and dihydroartemisinin-piperaquine (DHA-PPQ) that are deployed in the public sector. The status quo corresponds to continued use of AL and has the highest median monthly treatment failure count at 820,898. MFT strategies are color-coded based on AL usage: sky blue represents strategies with no AL; orange shows strategies where AL represents 25% to 50% of treatments; burnt orange shows strategies where AL is used for 75% of malaria cases. The vertical dashed grey line represents the median monthly treatment failures observed for a first-line switch to ASAQ. The MFT strategy with the lowest average monthly treatment failures is ASAQ (75%) - DHA-PPQ (25%), achieving a median monthly treatment failure count of 521,649, a 36.5% reduction compared to the status quo. Boxplot centers are medians, box bounds show interquartile range, and whiskers show 1.5 IQR.

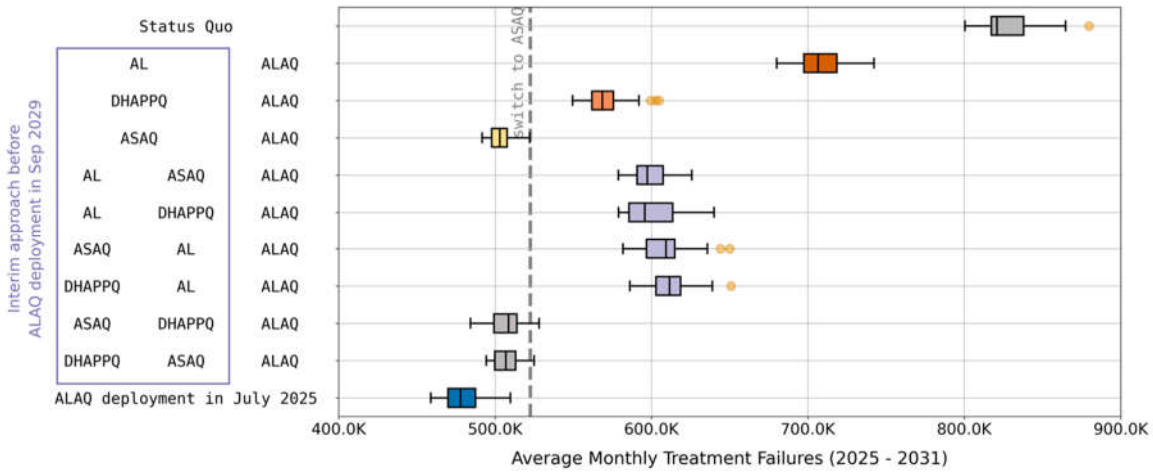

Figure 6. Average monthly treatment failures (2025-2031) under different triple ACT (ALAQ) deployment scenarios. Boxplots summarize N = 32 simulation outcomes and show the average monthly treatment failure counts for the period 2025-2031 for various ALAQ deployment scenarios. The vertical dashed grey line represents the median monthly treatment failures observed when ASAQ is used as first-line therapy after July 2025. If ALAQ is not available until July 2029, the best interim approach is to switch to ASAQ for four years or implement a 2-year cycle of ASAQ and DHA-PPQ before transitioning to ALAQ. These strategies effectively balance resistance management and treatment efficacy, with projected monthly treatment failures averaging ranging from 503,000 to 508,000. These represent substantial reductions in treatment failure counts compared to the status quo. In contrast, continuing AL use for four years before introducing ALAQ results in the worst outcomes, with monthly treatment failures reaching 706,494 (90% range: 687,993 – 736,851). This comparison highlights the effectiveness of an immediate switch to ALAQ and underscores the advantage of minimizing AL use as an interim strategy when ALAQ deployment is delayed. Boxplot centers are medians, box bounds show interquartile range, and whiskers show 1.5 IQR.

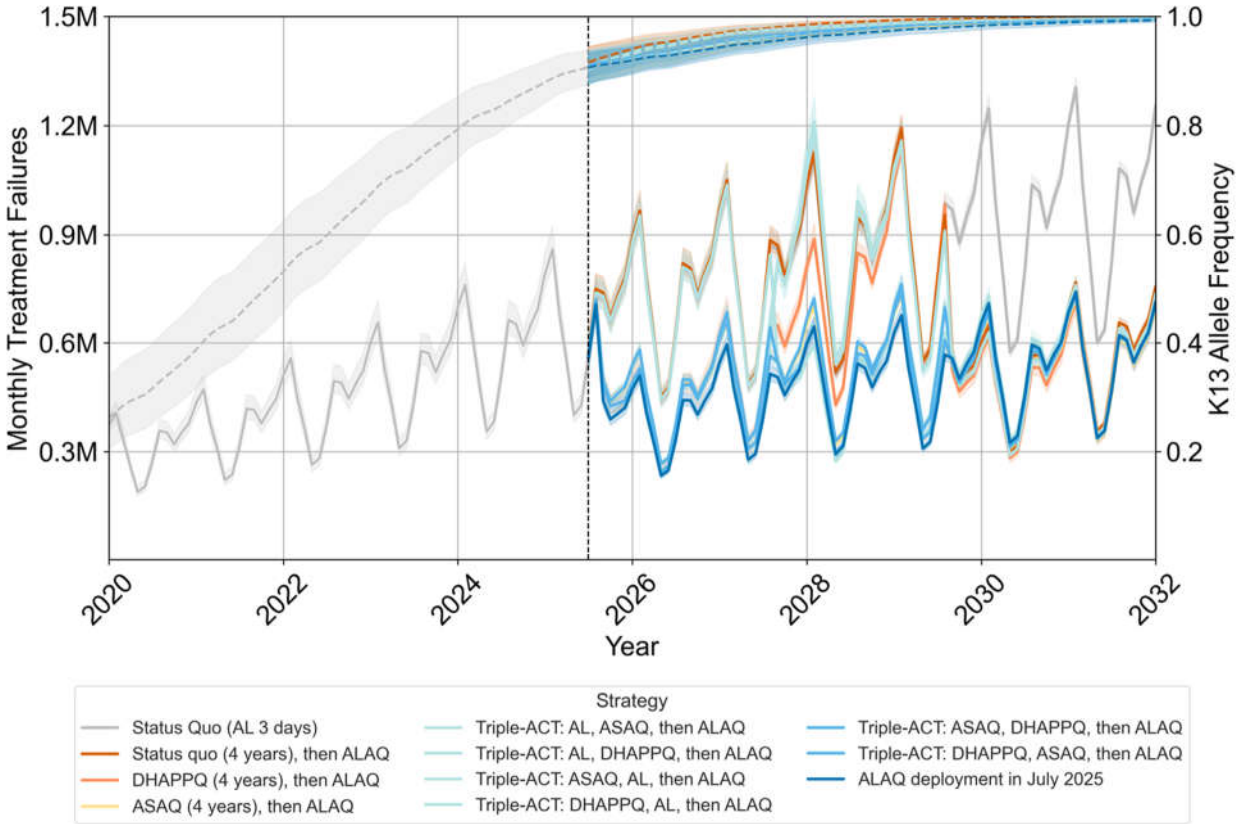

Figure 7. Trends in malaria treatment failures and *pfkelch13* allele frequency under different triple ACT (ALAQ) deployment scenarios through 2031. Gray lines show the model burn-in and calibration prior to July 2025. Solid lines show monthly treatment failure counts (left axis) while dashed lines show the combined *pfkelch13* allele frequency (sum of frequencies for the 469Y and 675V alleles). The monthly treatment failure count includes all projected treatment failures for all antimalarials used in the general population. Lines show medians while shaded regions denote the 90% ranges across simulation outcomes (N = 32). Blue line corresponds to immediate deployment of ALAQ, which achieves the best outcomes by keeping the average monthly treatment failure count below 500,000 and somewhat slowing the fixation kelch mutations. The sky-blue lines show 2-year cycling approaches using ASAQ and DHA-PPQ before switching to ALAQ. The orange line (DHA-PPQ), yellow line (ASAQ), and burnt orange line (AL) represent scenarios where a single ACT is used for four years before transitioning to ALAQ.

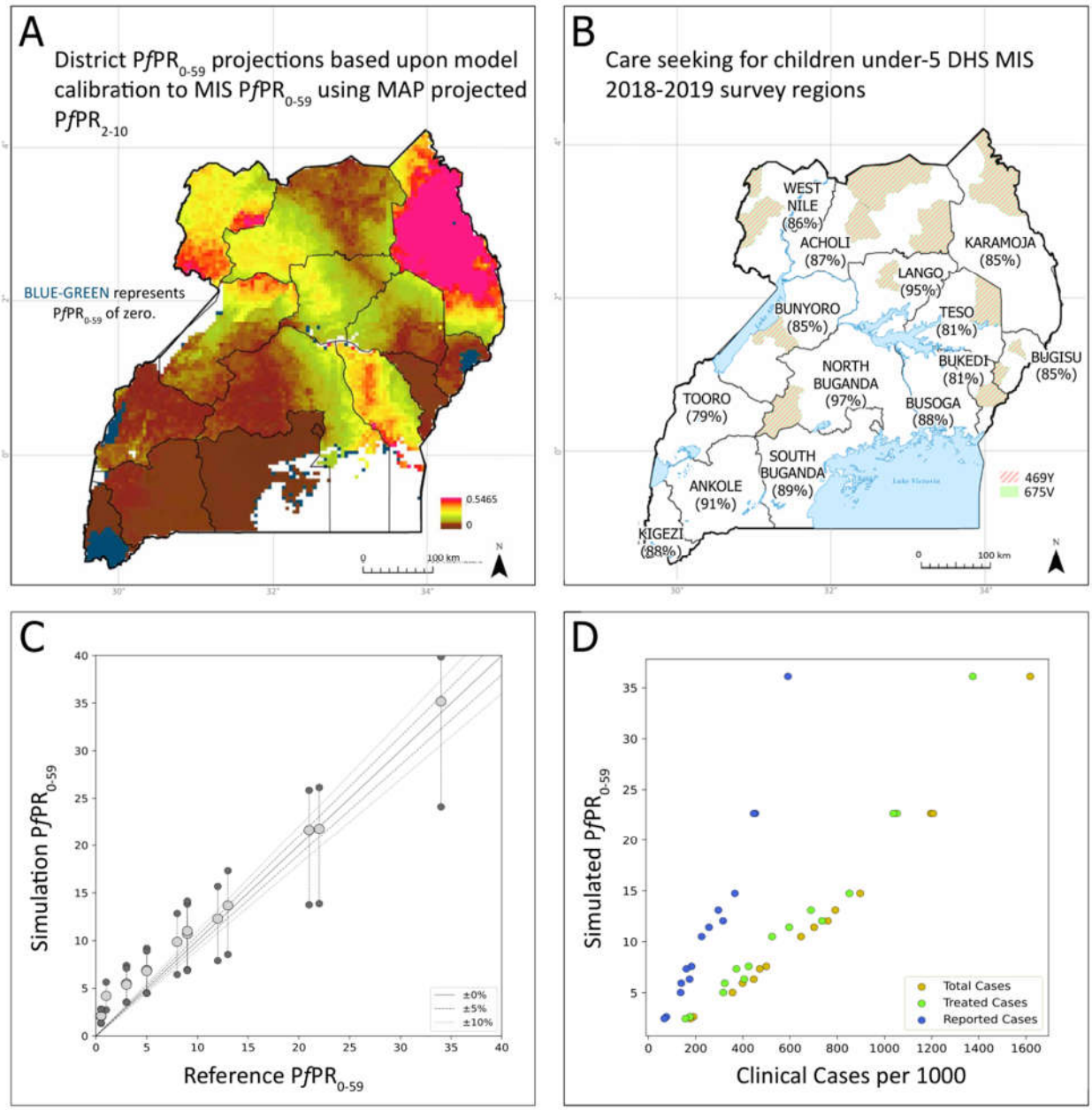
A
District PfPR0-59 projections based upon model calibration to MIS PfPR0-59 using MAP projected PfPR2-10
BLUE-GREEN represents PfPR0-59 of zero.
0.5465
0
0
100 km
N
B
Care seeking for children under-5 DHS MIS 2018-2019 survey regions
WEST NILE (86%)
ACHOLI (87%)
KARAMOJA (85%)
LANGO (95%)
BUNYORO (85%)
TESO (81%)
BUKEDI (81%)
BUGISU (85%)
NORTH BUGANDA (97%)
TOORO (79%)
BUSOGA (88%)
SOUTH BUGANDA (89%)
ANKOLE (91%)
KIGEZI (88%)
469Y
675V
0
100 km
N
C
Simulation PfPR0-59
Reference PfPR0-59
±0%
±5%
±10%
D
Simulated PfPR0-59
Clinical Cases per 1000
Total Cases
Treated Cases
Reported Cases

Figure 8. (A) Final prevalence ($PfPR_{0-59}$) calibration derived from Malaria Atlas Project estimates for $PfPR_{2-10}$ in 2017, and later adjusted to Malaria Indicator Survey (MIS) $PfPR_{0-59}$ values from 2018-2019. Map prepared by authors using simulation outputs, MIS 2018 DHS Regions Boundaries, and Uganda Boundaries. (B) Treatment coverage for children under five, according to 2018-2019 MIS. Shaded districts show districts where *pfkelch13* alleles 469Y or 675V have been identified. (C) Comparison of model-simulated prevalence in MIS districts to MIS data. The gray dot shows the annual mean prevalence in 0-5 year-olds while the dark gray dots at top and bottom indicate the intra-year upper and lower bounds. (D) By MIS district, the incidence of simulated malaria cases per year (tan), the incidence of simulated malaria cases that are treated (light green), and the incidence of treated cases that are likely to be reported to the public health system (blue) based on treatment seeking choices made between private and public sector. Incidence is shown as annual cases per 1000 individuals (x-axis) and plotted against the simulated $PfPR_{0-59}$ (y-axis).